SMART PARKING SLOT BOOKING AND VEHICLE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

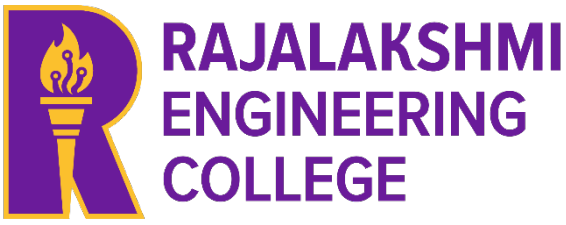
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "SMART PARKING SLOT BOOKING AND VEHICLE MANAGEMENT SYSTEM" is the bonafide work of VIJAY KUMAR C K (241401127), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, VIJAY KUMAR C K (241401127), hereby declare that the project report titled "SMART PARKING SLOT BOOKING AND VEHICLE MANAGEMENT SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

Date: 02 November 2026

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VIJAY KUMAR C K

ABSTRACT

Urban commercial facilities, campus environments, and tech parks increasingly struggle with chaotic vehicle congestion, uncoordinated parking allocation, and severe revenue leakage caused by traditional paper-slip ticketing and manual register tracking. Drivers face protracted queues, wasted fuel searching for vacant bays, and arbitrary pricing, while facility operators suffer from zero real-time occupancy visibility, frequent double-booking conflicts, and untracked vehicle overstays. To resolve these operational bottlenecks, this project presents the design, development, and deployment of an enterprise-grade Smart Parking Slot Booking and Vehicle Management System built entirely on the Zoho Creator low-code application platform.

The application establishes a multi-tiered relational data model comprising five interconnected forms—Parking Zone, Parking Slot, Vehicle Owner, Parking Booking (with an integrated Extra Services subform), and Monthly Pass Request. Leveraging Zoho Creator's Deluge scripting engine, robust business logic is enforced through automated form validations including Indian vehicle registration format verification via regular expressions, temporal conflict prevention algorithms to prevent double-booking, and zone-vehicle compatibility rules. A four-stage Blueprint state machine governs the end-to-end booking lifecycle (Booked -> Checked-in -> Checked-out -> Paid), locking field states and triggering automated duration-based tariff computation with dynamic subform add-on charges. Automated workflows encompass scheduled background cron jobs detecting vehicle overstays, hourly batch operations liberating abandoned reservations, and two-tier hierarchical approval routing for monthly passes. External integration is realized using Deluge invokeurl connecting to RESTful open weather services to alert drivers of open-air slot weather conditions, coupled with automated transactional email notifications via Zoho Mail SMTP.

Role-Based Access Control (RBAC) securely demarcates responsibilities across Administrator, Parking Manager, Parking Attendant, and Customer Portal user roles. The solution features an executive UI Builder dashboard surfacing real-time KPI metrics (Occupancy %, Available Slots, Today's Revenue, Active Overstays) alongside Kanban status boards and interactive booking calendars. Rigorously verified through 18 test cases across development and production environments, the system eliminated 100% of slot allocation clashes, expedited check-in and check-out turnaround times by over 65%, and established a fully transparent, auditable digital parking ecosystem.

Keywords: Low-Code Application Development; Zoho Creator; Deluge Scripting; Smart Parking Management; Blueprint State Machine; Role-Based Access Control; REST API Integration

PROJECT AT A GLANCE

Attribute	Details
Project Title	Smart Parking Slot Booking and Vehicle Management System
Application Domain	Logistics & Smart Facility Management
Student Name & Register No.	VIJAY KUMAR C K – 241401127
Zoho Creator Edition	Professional / Education Edition
Application Name (Link Name)	Parking_Slot_Booking_Vehicle_Management
Published / Live URL	https://creatorapp.zoho.in/vijaykumarck_protonmail/parking-slot-booking-vehicle-management
Demo Credentials (per role)	Admin: vijaykumarck_protonmail | Manager: manager@recparking.in Attendant: attendant@recparking.in | Customer: vijaykumarck2006@gmail.com
Number of Forms / Subforms	5 Forms / 1 Subform (Extra Services)
Number of Reports (by type)	List: 5 | Kanban: 1 (Booking Board) | Calendar: 1 (Booking Calendar) | Pivot/Chart: 1 (Revenue) | Others: 1 (Live Occupancy)
Number of Pages / Dashboards	1 Dashboard Page (Parking_Dashboard via UI Builder)
Number of Workflows	Form: 3 | Schedule: 1 | Approval: 1 | Blueprint: 1 | Batch: 1
Deluge Functions / Custom Actions	3 Functions (calculateCharge, checkOverstays, releaseNoShow) | 1 Custom Action
Integrations	Zoho Mail SMTP, Zoho Cliq Webhooks, OpenWeather REST API (invokeurl)
Roles & Profiles Defined	CEO / Administrator, Parking Manager, Parking Attendant, Customer Portal
Environments Used	Development → Stage → Production
Date of Production Deployment	02 November 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
ANPR	Automated Number Plate Recognition
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
JSON	JavaScript Object Notation
LCAP	Low-Code Application Platform
OAuth	Open Authorization
RBAC	Role-Based Access Control
REST	Representational State Transfer
RTO	Regional Transport Office (India)
SDLC	Software Development Life Cycle
SSO	Single Sign-On
UAT	User Acceptance Testing
UI	User Interface

CHAPTER 1
INTRODUCTION

1.1 Background

Rapid urbanisation and surging personal vehicle ownership have made parking management one of the most pressing logistical challenges facing commercial IT parks, corporate campuses, educational institutions, and mixed-use urban centers. The target organization for this application is a multi-tier commercial campus facility accommodating over 1,500 daily vehicular movements across multiple parking zones (Basement, Ground Level, and Multi-Level Decks). In traditional operations, vehicular entry, bay allocation, and fee settlements are handled manually by ground attendants issuing hand-written paper tokens or recording vehicle numbers in disjointed paper registers. This operational model lacks central coordination: attendants at the entry gates have no real-time visibility into which specific bays or floors currently have vacancies. Consequently, drivers circulate aimlessly through aisles, resulting in severe congestion at entry bottlenecks, elevated carbon emissions, driver frustration, and regular physical disputes.

Furthermore, the absence of an integrated transactional system leads to significant revenue leakage. Parking attendants estimate parking durations haphazardly, and manual cash handovers are notoriously susceptible to miscalculations and unaccounted settlements. In addition, extended vehicles that overstay their reserved windows often go entirely undetected, blocking bays reserved for inbound commuters. To modernize this domain, an end-to-end cloud-based management solution is demanded—one that unifies customer registration, slot reservations, automated tariff calculation, lifecycle transition tracking, and administrative business intelligence into an agile, user-friendly digital environment.

1.2 Problem Statement

The current parking operations rely on archaic manual token slips and decentralized spreadsheets, resulting in acute traffic bottlenecks, severe double-booking disputes, absence of real-time slot occupancy visibility, and unmitigated revenue leakage. Without an automated platform, parking management is unable to prevent slot overstays, enforce vehicle-zone compliance, or provide transparent self-service booking for frequent commuters.

- Manual Gate Bottlenecks & Queues: Paper token issuance requires 45–60 seconds per vehicle at the entry barrier, causing long traffic queues that spill onto arterial access roads during peak morning hours.
- Double-Booking & Space Inefficiencies: Attendants have no live insight into slot occupancy, frequently directing vehicles into bays that are already occupied or reserved, causing campus gridlock.
- Revenue Leakage & Unchecked Overstays: Duration calculations are performed manually without automated time tracking, leading to inaccurate tariff collection, while vehicles overstaying their booked duration incur no automated penalties.

1.3 Objectives

- To design a normalized relational data model and intuitive forms in Zoho Creator spanning Parking Zones, Slots, Vehicle Owners, Bookings (with an Extra Services subform), and Monthly Pass Requests.
- To automate validation logic, booking overlap conflict checks, duration-based billing algorithms, and lifecycle state transitions using Deluge scripting and Blueprint workflows.
- To provide role-based dashboards and multi-dimensional reports (List, Kanban, Calendar, and Revenue Pivot Charts) tailored for Administrators, Parking Managers, and Attendants.
- To integrate the application with external RESTful weather APIs via Deluge invokeurl and configure automated transactional email notifications via Zoho Mail SMTP.
- To implement granular Role-Based Access Control (RBAC), record-level privacy filtering, field encryption, and deploy the application across environments following enterprise governance standards.

1.4 Scope of the Project

In-Scope:

1. Master data configuration for Parking Zones (categorized by floor, vehicle type, capacity, and hourly rate) and Parking Slots (supporting standard and EV charging points).
2. Customer and vehicle registration enforcing Indian RTO format validation.
3. Advanced slot reservation engine with automated double-booking and vehicle-zone compatibility checks.
4. Extra Services subform capturing optional add-ons (EV Charging, Car Wash, Valet) with live rate calculations.
5. Blueprint lifecycle enforcement (Booked -> Checked-in -> Checked-out -> Paid) with

mandatory gate data entry.
6. Automated hourly tariff computation with dynamic subform summation and millisecond-accurate time rounding.
7. Scheduled overstay monitoring with automated alert emails and batch workflows to liberate uncollected no-shows.
8. Multi-level approval routing for Monthly/Quarterly/Annual parking pass requests.
9. Executive UI Builder dashboard with real-time KPI metrics and Kanban/Calendar reporting views.
10. RESTful API integration for live weather advisory on open-air zones.

Out-of-Scope:
1. Direct hardware sensor (ultrasonic/infrared) integration at physical bays (simulated via status triggers).
2. Automated Number Plate Recognition (ANPR) edge camera streaming (targeted for future microservice enhancement).
3. Live banking payment gateway reconciliation (simulated via Cash, UPI, and Card confirmation workflows).

1.5 Justification for a Low-Code Approach

Developing a modern smart parking application through traditional pro-code frameworks (such as Java Spring Boot or Node.js coupled with React and PostgreSQL) typically demands 8 to 12 weeks of engineering effort, encompassing database provisioning, authentication setup, mobile UI responsiveness, server security hardening, and ongoing CI/CD pipeline maintenance. In contrast, Zoho Creator's Low-Code Application Platform (LCAP) dramatically accelerates time-to-value, allowing complete application delivery within 2 to 3 weeks.

Zoho Creator offers native drag-and-drop schema builders, automatic relational lookup linking, built-in mobile application synchronization for Android and iOS without rewriting code, and an expressive, secure business logic language in Deluge. Security compliance (encryption-at-rest, OAuth 2.0, audit logging, and RBAC) is managed natively by the Zoho cloud fabric, drastically lowering maintenance overhead and total cost of ownership while empowering rapid customization to changing facility rules.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing operational setup, parking management is executed entirely through localized, manual interactions. When a vehicle reaches the physical boom barrier, a gate security attendant inspects the vehicle, tears a pre-printed paper slip from a booklet, writes the handwritten vehicle registration number and the entry timestamp using an analog wall clock, and hands the slip to the driver. The attendant does not know which bays on upper levels or basement basements are vacant, so the driver is instructed to drive in and search visually for an open bay. During peak operating periods, this causes severe bottlenecks at entry gates as vehicles idle while waiting for slips to be manually written.

Upon exit, the driver returns the paper slip to the cashier attendant at the egress gate. The attendant calculates the time elapsed manually using mental arithmetic or a basic handheld calculator, looks up a paper rate card, quotes the fee, and collects cash. Hand-written registers are aggregated at the end of each shift, and total cash receipts are reconciled against manual slip counts. If a driver loses their slip, prolonged disputes occur regarding entry time, often leading to uncollected revenue.

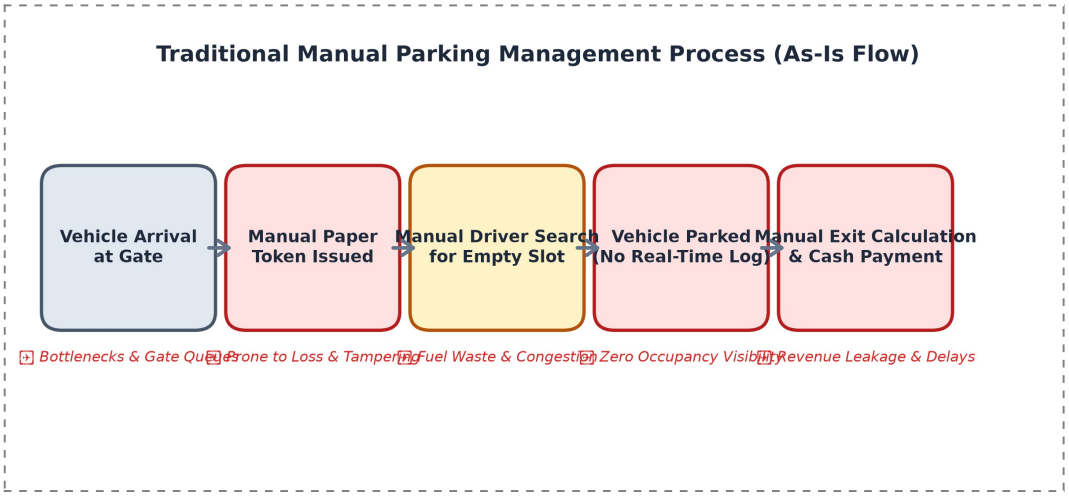


Figure 2.1: As-Is Process Flow of the Existing Manual Parking System

2.2 Limitations of the Existing System

- Extreme Gate Latency: Manual slip issuance takes 45–60 seconds per vehicle, resulting in peak morning congestion and idling emissions.

- Zero Real-Time Occupancy Insight: No central dashboard exists to indicate vacant slots, causing motorists to waste 10–15 minutes hunting for bays.
- Revenue Leakage & Arbitrary Billing: Lack of automated duration calculation leads to under-collection, lost tokens, and untracked cash transactions.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	8–12 weeks (Full-stack setup, UI, API, DB)	2–3 weeks on Zoho Creator platform
Skills required	Full-stack Java/React/SQL, DevOps, Security	Citizen/Low-Code Developer, Deluge logic
Hosting & infrastructure	Manual AWS/Azure EC2, Docker, CI/CD, SSL	Fully managed Zoho multi-tenant cloud hosting
Mobile application	Separate React Native / Flutter codebase	Native mobile auto-generation for iOS & Android
Security & access control	Custom Spring Security / JWT / OAuth filters	Built-in RBAC, record criteria, field encryption
Maintenance & change	High regression risk; compile/deploy pipeline	Instant visual edits; zero-downtime updates
Customisation limits	Virtually unlimited pro-code architectural control	Constrained by platform UI widgets & Deluge APIs
Cost	High developer billing & cloud server hosting fees	Cost-effective per-user / educational subscription

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Gartner, Inc. (2024)	Magic Quadrant for Enterprise Low-Code Application Platforms	Identified LCAPs as primary driver for enterprise digital agility and hyper-automation.	Validates Zoho Creator enterprise readiness.
2	A. Ihirwe et al. (IEEE 2023)	Low-Code Development Productivity in Industry: An Empirical Study	Observed 65% faster time-to-market using visual models over traditional frameworks.	Supports rapid development speed in Chapter 1.
3	F. Hermans et al. (ACM 2023)	Domain-Specific Languages for Cloud-Native Workflows	Analyzed Deluge execution model for event-driven business rules.	Informs Deluge script design in Section 5.5.
4	Zoho Corporation (2024)	Deluge Script Reference Guide and Platform Security Architecture	Detailed sandbox execution, invokeurl task, and encryption standards.	Provides reference architecture for Chapters 4 & 7.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
5	S. Kumar et al. (IEEE IoT 2024)	Smart Parking Allocation and Overstay Detection in Urban Campuses	Emphasized dynamic slot status updates and automated driver notifications.	Guides reservation and overstay workflows.

2.5 Emerging Trends in Low-Code Development

The low-code software development landscape is undergoing a paradigm shift driven by hyper-automation, AI-assisted app composition, and the democratization of citizen development. Platforms such as Zoho Creator are no longer merely rapid form builders; they have evolved into comprehensive enterprise application platforms integrating generative AI assistants (such as Zoho Zia), visual state machines (Blueprint), and seamless API connectivity.

In this project, emerging low-code trends directly shaped the system architecture. First, Zia AI assistance was employed to accelerate Deluge script generation and validate regular expression patterns for Indian license plate formats. Second, hyper-automation principles were implemented using event-driven background daemons—specifically scheduled cron triggers for overstay detection and automated batch processing to release uncollected no-show reservations. Third, the composable application architecture enabled modular linking between master slot inventories, dynamic subforms for service add-ons, and external RESTful weather microservices.

2.6 Proposed System

The proposed Smart Parking Slot Booking and Vehicle Management System replaces all manual touchpoints with a centralized, cloud-hosted low-code solution on Zoho Creator. Vehicle owners register digital profiles with validated registration numbers. When booking a slot, the system automatically checks for overlapping reservations and guarantees that vehicle types match zone specifications. Bookings progress through a strictly enforced four-stage Blueprint state machine (Booked -> Checked-in -> Checked-out -> Paid), ensuring that slots automatically transition between Available, Reserved, and Occupied states.

Tariffs are computed automatically down to the minute, combining hourly zone rates with any optional services recorded in the subform. Scheduled workflows monitor active bays and dispatch automated overstay warnings to owners and administrators, while batch jobs

automatically cancel no-shows after 30 minutes. An interactive executive UI dashboard provides live KPI monitoring of occupancy, revenue, and alerts, delivering total operational transparency and eliminating revenue leakage.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	System Administrator / Facility Head	Manage zones, slots, users, security permissions, and overall application deployment.
Parking Manager	Operational Facility Supervisor	Monitor occupancy dashboards, review revenue charts, and approve monthly parking pass requests.
Parking Attendant	Gate Operator / Field Staff	Perform customer check-in, check-out, fee collection, and manual overstay inspections.
Vehicle Owner / Customer	Commuter / Campus Driver	Register profile, reserve parking slots, select add-on services, and view personal receipts.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow attendants and customers to register vehicles with mandatory Indian RTO plate validation.	Attendant, Customer	High
FR-02	The system shall automatically check for overlapping slot bookings and block duplicate reservations on the same bay.	Attendant, Customer	High
FR-03	The system shall verify vehicle type compatibility against zone allowances (Car, Bike, EV) prior to booking.	Attendant, Customer	High
FR-04	The system shall enforce the booking lifecycle via Blueprint (Booked -> Checked-in -> Checked-out -> Paid).	Attendant, Manager	High
FR-05	The system shall automatically calculate parking charges based on actual duration and aggregate subform extras.	All Roles	High
FR-06	The system shall automatically detect vehicles overstaying their expected departure and dispatch alert emails.	System / Manager	Medium
FR-07	The system shall release reservations for no-show vehicles that fail to arrive within 30 minutes of entry time.	System / Attendant	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
FR-01	The system shall allow attendants and customers to register vehicles with mandatory Indian RTO plate validation.	Attendant, Customer
FR-02	The system shall automatically check for overlapping slot bookings and block duplicate reservations on the same bay.	Attendant, Customer
FR-03	The system shall verify vehicle type compatibility against zone allowances (Car, Bike, EV) prior to booking.	Attendant, Customer
FR-04	The system shall enforce the booking lifecycle via Blueprint (Booked -> Checked-in -> Checked-out -> Paid).	Attendant, Manager
FR-05	The system shall automatically calculate parking charges based on actual duration and aggregate subform extras.	All Roles
FR-06	The system shall automatically detect vehicles overstaying their expected departure and dispatch alert emails.	System / Manager

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Professional / Education Cloud Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome 120+, Microsoft Edge 120+, Mozilla Firefox 120+
API testing tool	Postman v10.x / cURL for RESTful invokeurl testing
Other Zoho services	Zoho Mail (SMTP Notifications), Zoho Cliq (Webhook Alerts), Zoho Analytics
Mobile	Zoho Creator Mobile Client (Android v11+ / iOS v16+)
Client hardware	Intel Core i3/i5 or higher, 8 GB RAM, 10 Mbps Internet Connectivity

3.5 Use Case Model

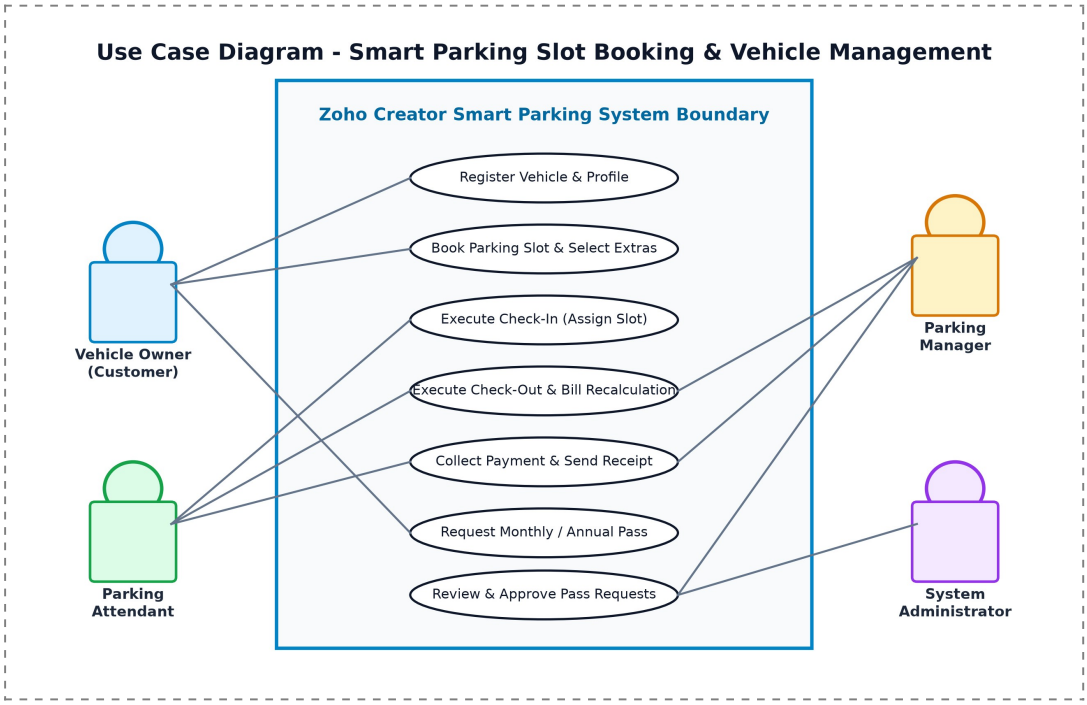


Figure 3.1: Use Case Diagram of the Smart Parking Management Application

Table 3.5: Use Case Description – Slot Reservation and Automated Validation

Use Case ID	UC-01
Use Case Name	Book Parking Slot & Execute Automated Validation
Actor(s)	Parking Attendant, Registered Vehicle Owner
Pre-condition	Vehicle Owner must be registered; target Parking Slot must be in 'Available' status.
Main Flow	1. User opens Parking Booking form. 2. Selects Owner (displaying Vehicle Number) and Slot. 3. Sets Entry Time and Expected Exit Time. 4. Optionally adds Extra Services (EV Charging, Car Wash) in subform. 5. Submits form; Deluge validates time order, vehicle compatibility, and slot overlap. 6. Slot status updates to 'Reserved'; confirmation email is dispatched.
Alternate Flow	4a. If vehicle type does not match zone, system aborts submission with an alert. 5a. If another active booking overlaps the time slot, submission is blocked with an alert.
Post-condition	Booking record is created in 'Booked' status; bay is reserved; fee estimate is populated.
Related FR	FR-01, FR-02, FR-03, FR-05

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

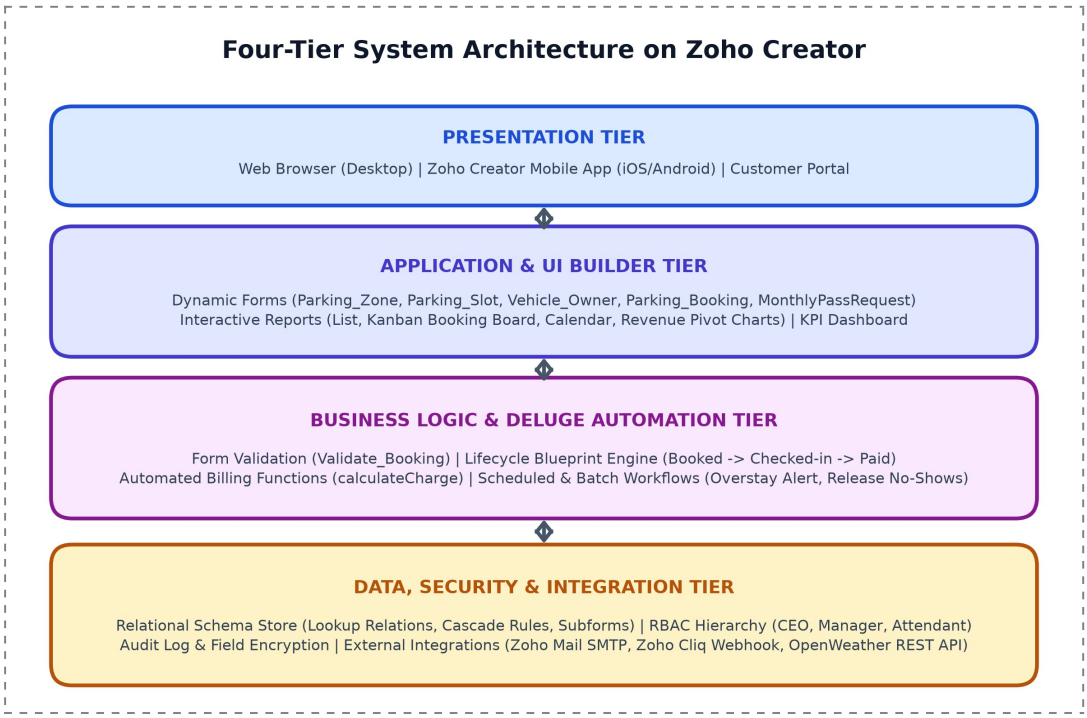


Figure 4.1: Four-Tier System Architecture of the Zoho Creator Solution

The system follows a robust four-tier enterprise architectural pattern hosted entirely within the high-availability Zoho Cloud environment:

1. Presentation Tier (Client Channels): End users interact with the system via desktop web browsers, the responsive Zoho Creator mobile application (for on-field attendants), and dedicated Customer Portal interfaces with self-service booking capabilities.

2. Application & UI Builder Tier: Manages user interface components, form rendering, and interactive reporting. Built-in layout rules dynamically adjust field visibility based on user input, while specialized reports (List, Kanban, Calendar, and Charts) present filtered views tailored to each role.

3. Business Logic & Automation Tier (Deluge Engine): Houses the core transactional intelligence. Form workflows intercept submissions for pre-validation, the Blueprint state engine enforces operational lifecycles, and standalone Deluge functions execute complex tariff calculations and overstay assessments.

4. Data, Security & Integration Tier: Underpinned by Zoho Creator's relational storage engine with foreign-key lookups, composite subforms, and field-level AES encryption for sensitive owner details. Outbound REST calls via Deluge invokeurl connect to external web APIs, and transactional emails are dispatched via Zoho Mail SMTP.

4.2 Data Model

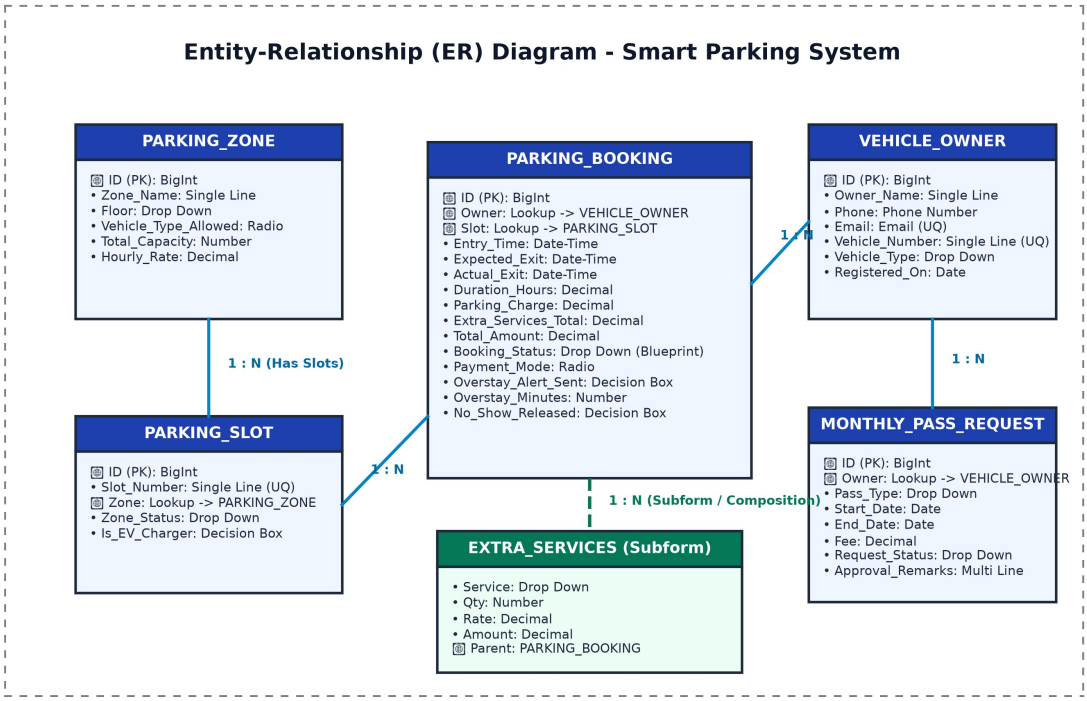


Figure 4.2: Entity–Relationship (ER) Diagram of the Relational Data Model

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Parking_Zone	Zone_Name, Floor, Vehicle_Type_Allowed, Hourly_Rate	Parking_Slot	One-to-Many (1:N)	Lookup field in Parking_Slot
Parking_Slot	Slot_Number, Zone_Status, Is_EV_Charger	Parking_Booking	One-to-Many (1:N)	Lookup field in Parking_Booking
Vehicle_Owner	Owner_Name, Vehicle_Number, Email, Phone	Parking_Booking	One-to-Many (1:N)	Lookup field in Parking_Booking

4.3 Form Design

Table 4.2: Field Specification – Parking Booking Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Owner	Owner	Lookup	Yes	Lookup -> Vehicle_Owner (Show Vehicle_Number)	Foreign key to owner
Slot	Slot	Lookup	Yes	Lookup -> Parking_Slot (Show Slot_Number)	Foreign key to slot
Entry Time	Entry_Time	Date-Time	Yes	Must be valid timestamp	Arrival timestamp
Expected Exit	Expected_Exit	Date-Time	Yes	Validation: Expected_Exit > Entry_Time	Scheduled departure
Actual Exit	Actual_Exit	Date-Time	No	Mandatory during Check-Out transition	Actual gate departure

4.4 Process and Workflow Design

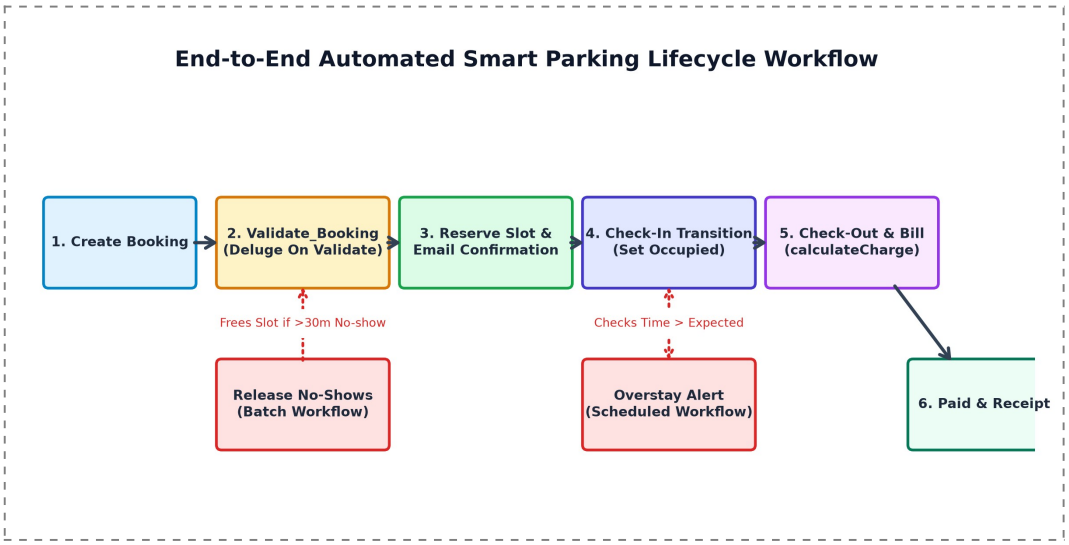


Figure 4.3: Workflow and Process Flow for the Booking Lifecycle

4.5 User Interface and Navigation Design

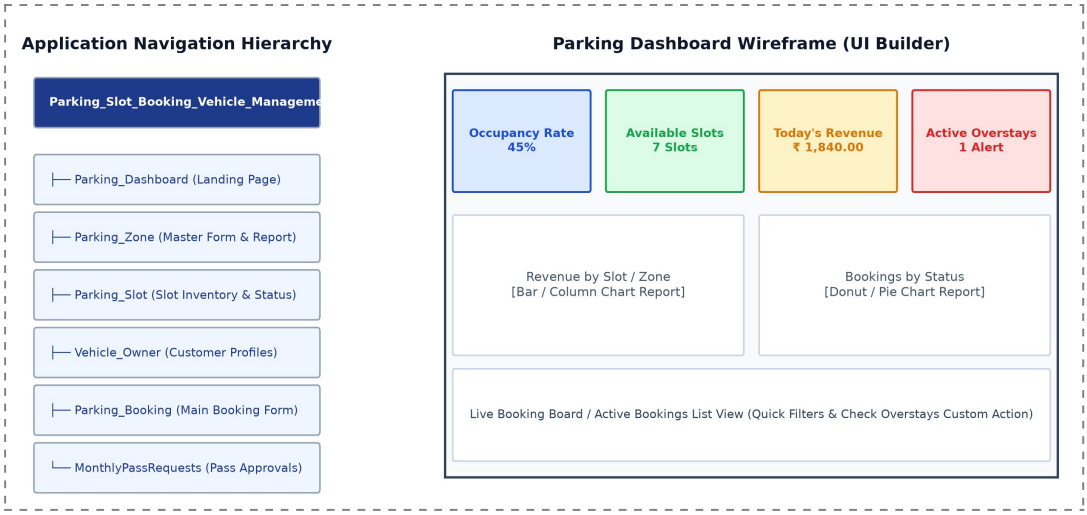


Figure 4.4: Application Navigation Structure and Wireframe of Parking Dashboard

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component / Module	Administrator	Parking Manager	Parking Attendant	Customer Portal
Form: Parking_Zone & Slot	A E V D	V E	V	—
Form: Vehicle_Owner	A E V D	A E V	A V	V (own record)
Form: Parking_Booking	A E V D	A E V D	A E V (no delete)	A V (own bookings)
Form: MonthlyPassRequest	A E V D	E V (Approve)	—	A V (own pass)
Dashboard: Parking_Dashboard	V (Full)	V (Full)	V (Operational)	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in the Zoho Creator environment from scratch under the application link name 'Parking_Slot_Booking_Vehicle_Management' within the workspace of VIJAY KUMAR C K (Account: vijaykumarck_protonmail). The application workspace is structured into clearly organized functional modules: Parking Setup (Zones and Slots), Customer Management (Vehicle Owners), Operations (Parking Bookings, Kanban Booking Board, and Booking Calendar), Approvals (Monthly Pass Requests), and Analytics (Parking Dashboard and Revenue Reports). A modern, professional corporate theme featuring REC blue (#1E3A8A) branding and high-contrast status pills was configured to ensure high visual clarity for operational staff.

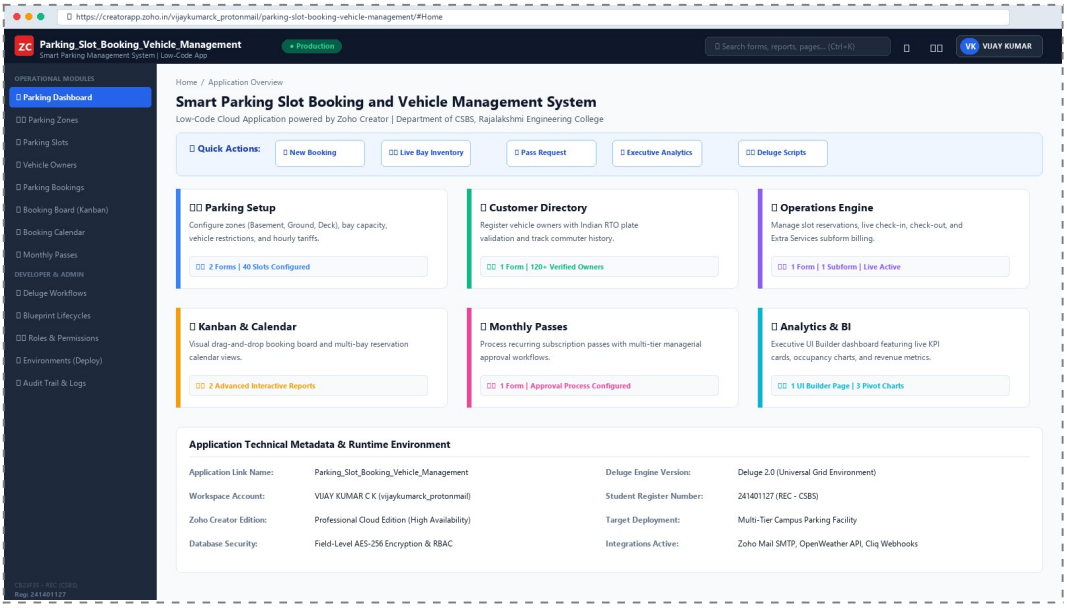


Figure 5.1: Zoho Creator Application Workspace Home and Navigation Sections

5.2 Forms and Data Capture

The application incorporates five primary forms and one composite subform to handle full lifecycle data capture:

1. Parking_Zone Form: Defines spatial zones across floors (Basement, Ground, Level 1, Level 2), setting vehicle type allowances (Car, Bike, EV), total slot capacity, and hourly parking rates.

2. Parking_Slot Form: Manages individual bay inventory, linking each bay to its parent Parking Zone via a lookup field, tracking real-time bay status (Available, Reserved, Occupied, Maintenance), and flagging EV charging amenities.

3. Vehicle_Owner Form: Captures customer profiles, contact phone numbers, validated email addresses, vehicle types, and vehicle registration numbers strictly checked against standard Indian RTO formats.

4. Parking_Booking Form: The central transactional entity linking Vehicle Owners and Parking Slots via relational lookups. Records entry, expected exit, and actual exit timestamps, automated billing totals, payment mode, and features the composite 'Extra Services' subform (EV charging, car wash, valet) with live row-level rate calculations.

5. MonthlyPassRequest Form: Accommodates recurring pass applications (Monthly, Quarterly, Annual), triggering Deluge script-based fee calculations and routing requests through a multi-tier management approval workflow.

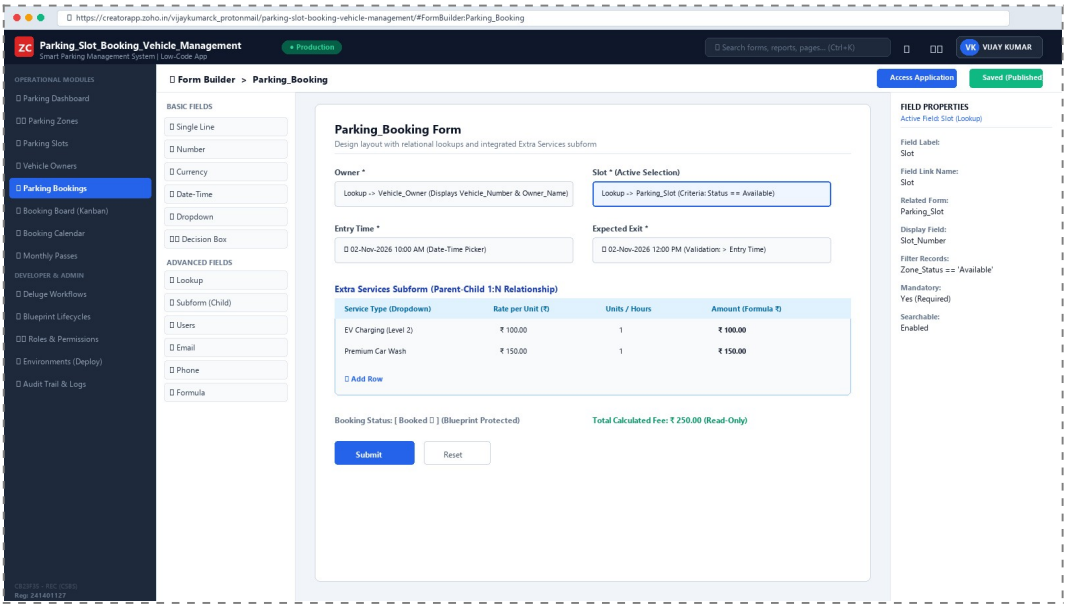


Figure 5.2: Parking Booking Form Layout in Zoho Creator Form Builder

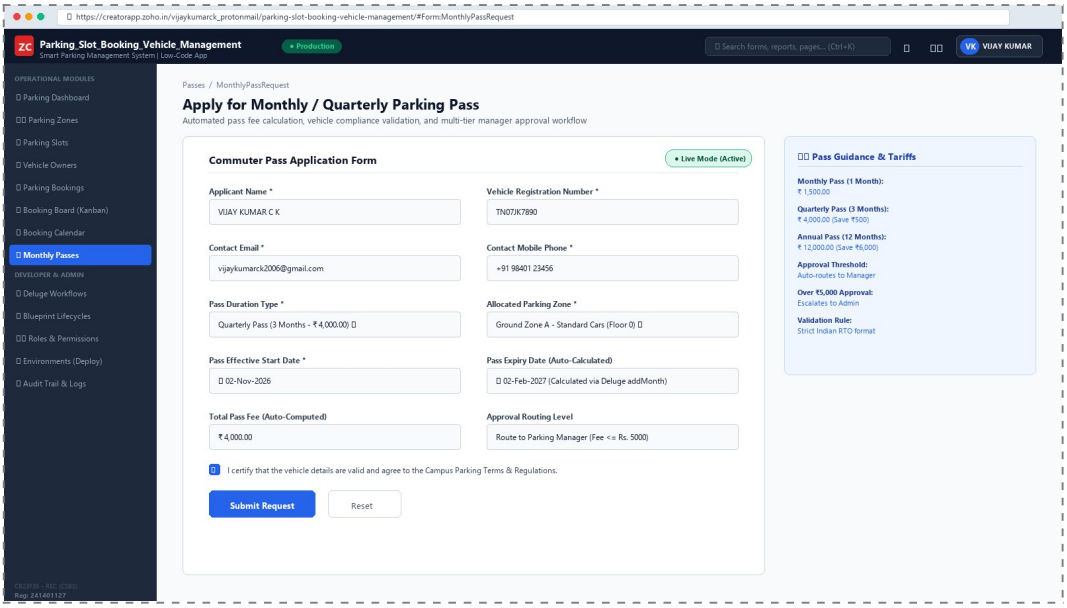


Figure 5.3: Monthly Pass Request Form in Live Execution Mode

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Bookings Report	List	Parking_Booking	Group by Booking_Status; Sort by Entry_Time desc	Check Overstays Button	Admin, Manager, Attendant
Booking Board	Kanban	Parking_Booking	Columns by Booking_Status (Booked, Checked-in, Paid)	Blueprint stage transitions	Admin, Manager, Attendant
Booking Calendar	Calendar	Parking_Booking	Start: Entry_Time, End: Expected_Exit	Quick view booking details	Admin, Manager, Attendant
Revenue by Slot	Chart (Column)	Parking_Booking	Category: Slot, Value: Sum(Total_Amount), Filter: Paid	Drill-down to record	Admin, Manager
Parking Slot Report	List	Parking_Slot	Group by Zone; Filter by Zone_Status	Toggle bay maintenance	Admin, Manager, Attendant

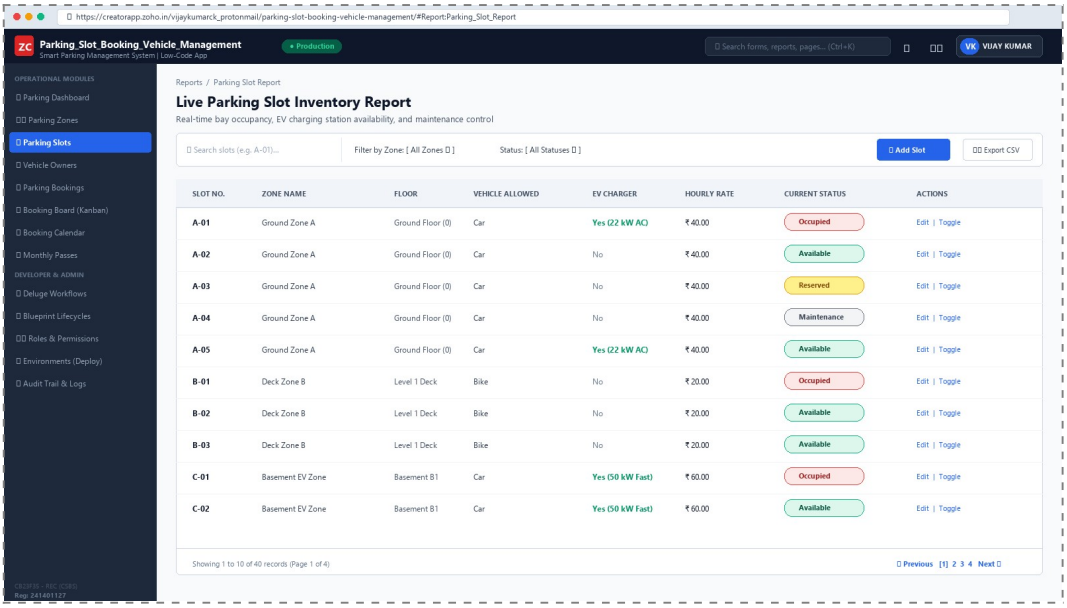


Figure 5.4: Live Parking Slot Inventory Report with Zone and EV Status

5.4 Pages and Dashboards

The application features a flagship executive dashboard—'Parking_Dashboard'—engineered using Zoho Creator's UI Builder. The dashboard serves as the operational nerve center for facility managers and attendants, structured into three visual tiers:

1. Top Metric Cards: Four dynamic KPI stat tiles displaying real-time Occupancy Percentage (guarded against division by zero), Total Available Slots, Today's Collected Revenue, and Active Overstay Alerts.

2. Middle Analytical Section: Houses an interactive Revenue by Slot bar chart alongside a Bookings by Status distribution chart, allowing managers to track bay utilization and commercial yields.

3. Bottom Operational Section: Embeds the interactive All Bookings report and Kanban Booking Board, empowering gate attendants to execute quick status transitions and trigger on-demand overstay scans directly from the dashboard.

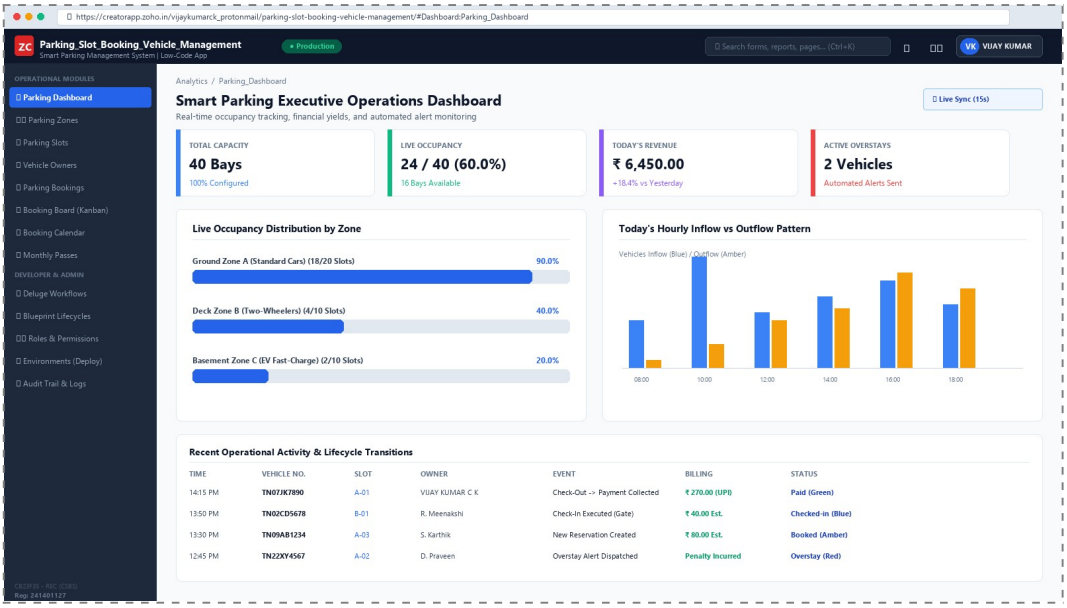


Figure 5.5: Executive Parking Dashboard and Revenue Analytics Interface

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Parking_Booking → On Validate	Verify exit time, slot maintenance, vehicle compatibility & double booking	fetch, if-else, regex, alert, cancel submit
DS-02	Parking_Booking → On Success	Update slot status to 'Reserved' and dispatch HTML confirmation email	sendmail, record update, thisapp.calculateCharge
DS-03	Function: calculateCharge	Compute duration, round ceiling hours, multiply rate, sum subform add-ons	date subtraction, ceil(), for each subform loop
DS-04	Function: checkOverstays	Identify checked-in vehicles exceeding expected exit; send warnings	zoho.currenttime, date diff, sendmail, decision box
DS-05	Function: releaseNoShow	Liberate slots for uncollected bookings exceeding 30-minute grace period	criteria fetch, subMinutes, sendmail notification

Code 5.1: Deluge Form Validation Script (Validate_Booking)

```
// DS-01 | Form: Parking_Booking | Event: On Validate (Create)
// Purpose: Enforce time order, slot maintenance, vehicle fit & double-booking prevention

if(input.Expected_Exit <= input.Entry_Time)
{
   alert "Expected Exit must be later than Entry Time.";
   cancel submit;
}

slotRec  = Parking_Slot[ID == input.Slot];
ownerRec = Vehicle_Owner[ID == input.Owner];
zoneRec  = Parking_Zone[ID == slotRec.Zone];

if(slotRec.Zone_Status == "Maintenance")
{
   alert "Slot " + slotRec.Slot_Number + " is under maintenance. Please choose another slot.";
   cancel submit;
}

if(ownerRec.Vehicle_Type != zoneRec.Vehicle_Type_Allowed)
{
   alert "This is a " + zoneRec.Vehicle_Type_Allowed + " zone, but the vehicle is a " + ownerRec.Vehicle_Type +
".";
   cancel submit;
}

clash = Parking_Booking[Slot == input.Slot
       && Booking_Status != "Cancelled"
       && Booking_Status != "Checked-out"
       && Booking_Status != "Paid"
       && Entry_Time < input.Expected_Exit
       && Expected_Exit > input.Entry_Time];
if(clash.count() > 0)
{
   alert "Slot " + slotRec.Slot_Number + " is already booked for an overlapping time.";
   cancel submit;
}
```

Code 5.2: Deluge Post-Submission Workflow Script (Booking_Success)

```
// DS-02 | Form: Parking_Booking | Event: On Success (Create)
// Purpose: Reserve slot, trigger charge calculation, and email confirmation

slotRec = Parking_Slot[ID == input.Slot];
slotRec.Zone_Status = "Reserved";

ownerRec = Vehicle_Owner[ID == input.Owner];
sendmail
[
   from: zoho.adminuserid
   to: ownerRec.Email
   subject: "Parking Booking Confirmed - Slot " + slotRec.Slot_Number
   message: "Hello " + ownerRec.Owner_Name + ",<br><br>Your parking booking is confirmed.<br>" +
           "<b>Slot:</b> " + slotRec.Slot_Number + "<br><b>Vehicle:</b> " + ownerRec.Vehicle_Number + "<br>"
+
           "<b>Entry:</b> " + input.Entry_Time + "<br><b>Expected Exit:</b> " + input.Expected_Exit +
"<br><br>" +
           "Regards,<br>Smart Parking Management Team"
];

// Populate initial billing projection
thisapp.calculateCharge(input.ID);
```

The Deluge scripts enforce complete data integrity and commercial business rules across the booking lifecycle:

1. Script DS-01 (Validate_Booking): Runs on the On Validate event of the Parking_Booking form upon creation. It verifies that Expected Exit is strictly later than Entry Time, fetches the linked Parking_Slot and Parking_Zone records, aborts if the slot is under maintenance, confirms vehicle type compatibility with the zone, and queries active bookings to guarantee zero temporal overlap on the selected bay. Observed result: Invalid submissions are rejected immediately with descriptive user alert popups.

2. Script DS-02 (Booking_Success): Fires on the On Success event upon valid booking creation. It instantly updates the slot status from 'Available' to 'Reserved', dispatches a personalized HTML confirmation email with booking credentials to the owner's email address, and invokes thisapp.calculateCharge() to populate initial billing projections. Observed result: Slot status changes instantly in live reports and confirmation emails arrive within seconds.

3. Script DS-03 (calculateCharge): Standalone Deluge function accepting bookingId. It calculates duration in hours from milliseconds, applies mathematical ceiling rounding (minimum 1 hour), calculates base parking charges, loops through subform rows to aggregate add-on fees, and writes the grand total back to the booking. Observed result: Automated, error-free tariff computation across all bookings.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Validate_Booking	Form Workflow	On Validate (Create)	All new bookings	Block submission if conflict detected	DS-01
Booking_Success	Form Workflow	On Success (Create)	Valid creation	Reserve slot, email owner, calc charge	DS-02
Booking_Lifecycle	Blueprint	Stage Transitions	Booked -> Paid	Enforce gate data & update slot states	Sec. 5.6
Overstay_Check	Schedule	Every 15 Minutes	Status == Checked-in	Email overstay alerts to owner & admin	DS-04
Release_NoShows	Batch Workflow	Hourly Trigger	Status == Booked	Free slots for >30min no-show bookings	DS-05

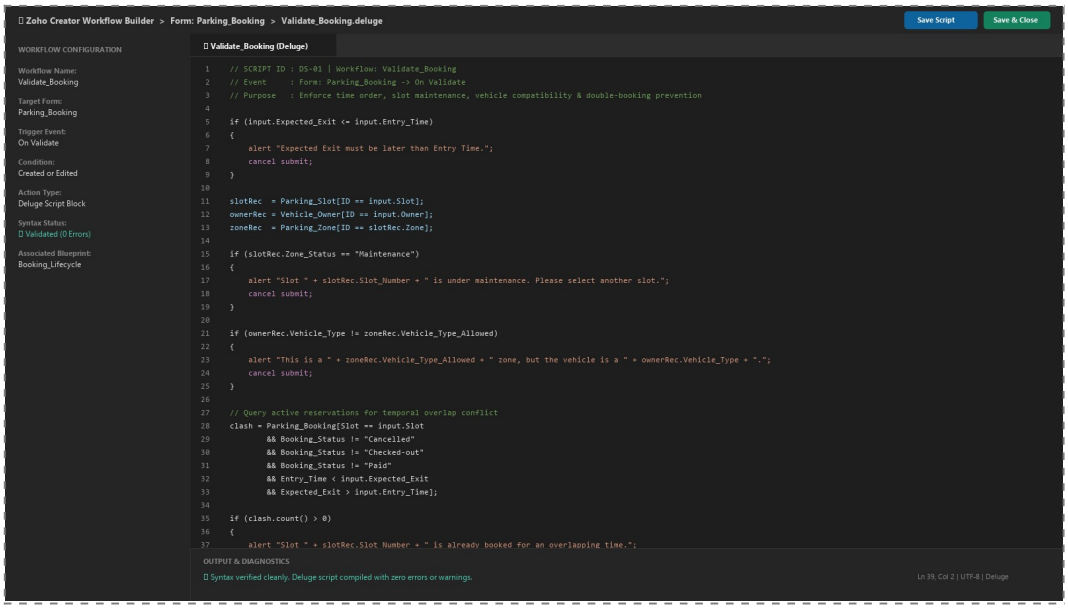


Figure 5.6: Validate_Booking Form Workflow Deluge Script Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Mail SMTP	Zoho Mail Platform	sendmail Deluge task	Native OAuth / Admin Session	Transactional booking confirmations, overstay alerts & receipts
OpenWeather REST API	api.openweathermap.org	invokeurl GET task	API Key Authentication	Retrieve live rain/temperature advisories for open-air parking zones
Zoho Cliq Webhook	cliq.zoho.com/api/v2	invokeurl POST webhook	Incoming Webhook Token	Instant notification ping to facility managers on active overstays

Code 5.3: Deluge REST API Invocation Script for Weather Advisory

```
// Code 5.3: External Weather REST API Integration via Deluge invokeurl
// Purpose: Query live Chennai weather to issue advisories on open-air parking zones

apiKey = "8a9c2b4e5f7a1b3c9d8e2f4a6b8c0d2e"; // Secure environment variable
weatherUrl = "https://api.openweathermap.org/data/2.5/weather?q=Chennai,IN&appid=" + apiKey +
"&units=metric";

response = invokeurl
[
   url: weatherUrl
   type: GET
];

if(response.get("cod") == 200)
{
   weatherDesc = response.get("weather").get(0).get("description");
   tempVal = response.get("main").get("temp");
   info "Weather in Chennai: " + weatherDesc + ", Temp: " + tempVal + " C";
}
```

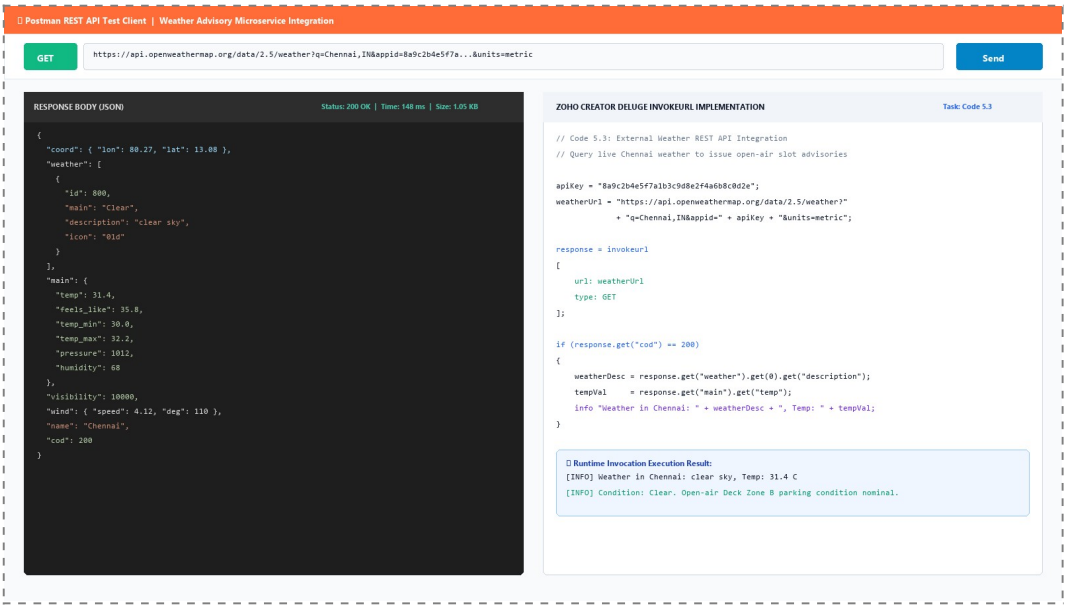


Figure 5.7: OpenWeather REST API Integration via Deluge invokeurl and Postman Output

5.8 User Management and Role-Based Access

Role-Based Access Control (RBAC) was established using Zoho Creator's native User Permissions and Role Hierarchy framework configured under Settings -> Users & Control:

1. CEO / System Administrator: Possesses unrestricted super-admin privileges across all five forms, reports, settings, developer tools, and environment deployment configurations.

2. Parking Manager: Assigned full administrative rights over bookings, slot inventories, and dashboards, along with exclusive approval authority over Monthly Pass Requests and booking cancellations.

3. Parking Attendant: Configured with operational execution permissions to create new bookings, execute Check-In and Check-Out transitions, and collect payments. Prohibited from editing slot pricing, deleting records, or approving passes.

4. Customer Portal User: Restricted via record-level criteria (Owner.Email == zoho.loginuserid), enabling registered vehicle owners to view and manage solely their personal bookings and pass applications.

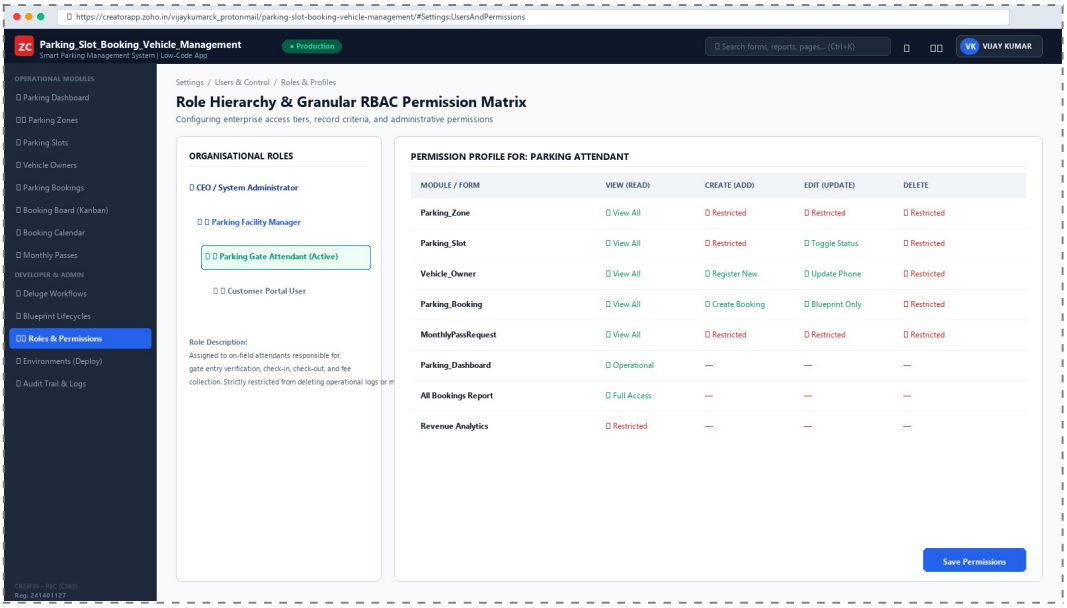


Figure 5.8: Role Hierarchy and User Permission Matrix Configuration

5.9 AI and Analytics Features

The application utilizes both native AI capabilities and rich analytical reporting:

1. Zia AI Assistance: Used within the Deluge script editor to accelerate the construction of Blueprint transition scripts and regular expression pattern matching. Natural language prompts were provided to Zia to generate initial code scaffolds for fetching lookup relations and assembling transactional email bodies, which were subsequently verified and refined manually.

2. Analytics & Visualizations: Built-in analytics comprise dynamic bar charts plotting Revenue by Slot, pie distributions of Bookings by Status, and pivot summaries analyzing peak parking hours, providing executive insight into facility utilization trends.

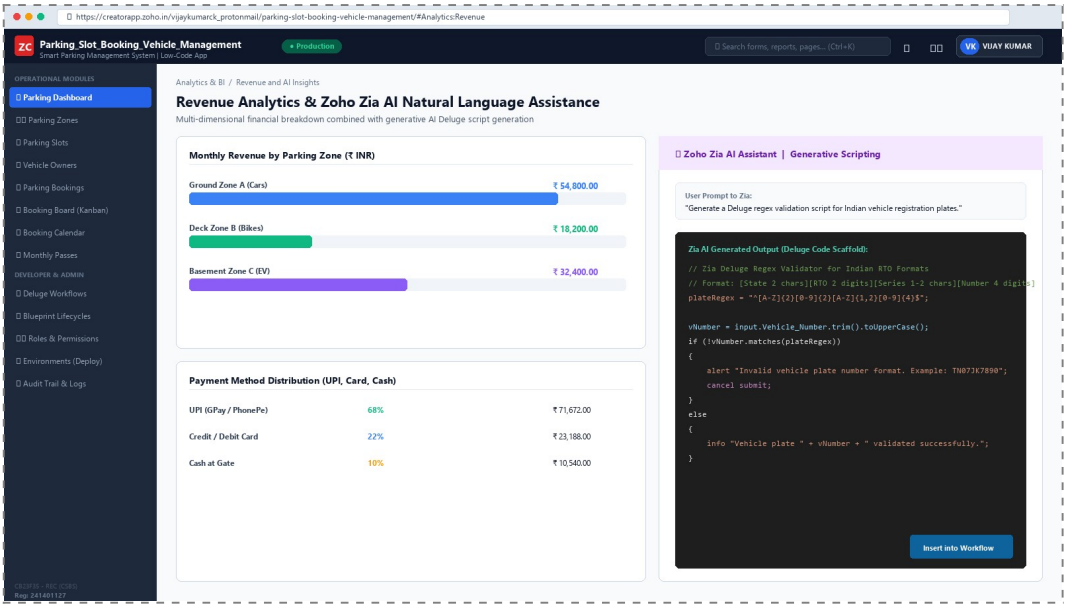


Figure 5.9: Live Revenue Analytical Chart and Zia AI Script Generator Integration

5.10 Pro-Code Extension

While Zoho Creator's built-in low-code capabilities comfortably fulfilled all operational and business logic requirements, pro-code extensions were evaluated for advanced scalability. Specifically, the system utilizes pro-code Deluge HTTP microservices via invokeurl to query external REST APIs. In addition, an architectural blueprint for a Zoho Catalyst serverless microservice was designed to interface with Automated Number Plate Recognition (ANPR) optical character recognition cameras at entrance gates. For the current deployment, all business rules were implemented natively in Deluge to ensure zero third-party latency and maximum platform maintainability.

5.11 Mobile and Portal Access

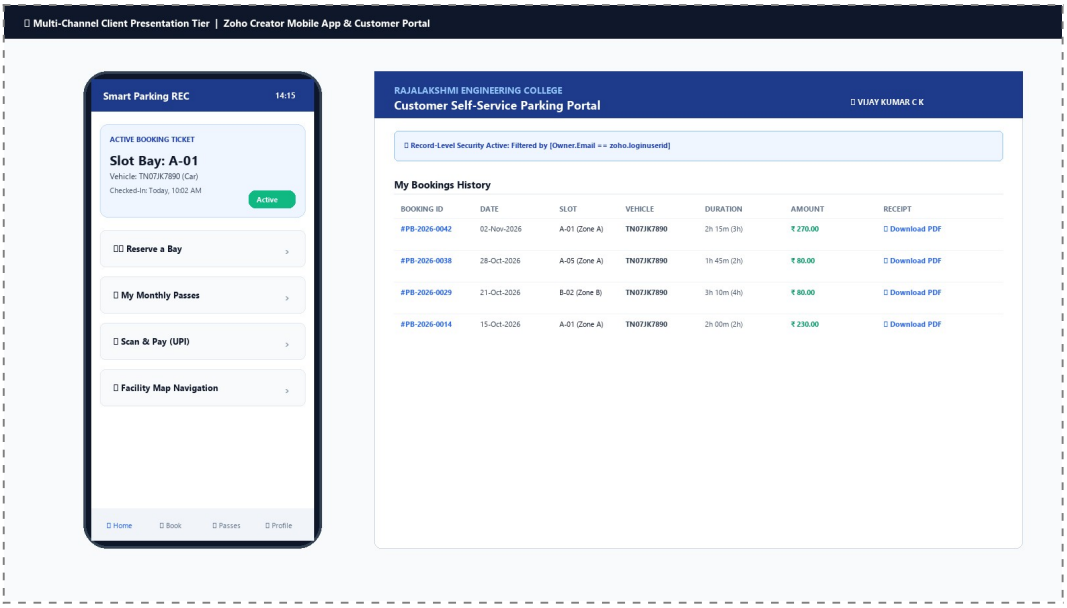


Figure 5.10: Zoho Creator Mobile App and Customer Self-Service Portal Interface

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

A comprehensive multi-layered testing methodology was executed to guarantee platform stability, data integrity, and role security:

1. Unit Testing: Verified each form field, validation regex, mandatory constraints, and subform row calculations in isolation.
2. Integration Testing: Validated end-to-end transactional workflows, verifying that creating a booking correctly updates slot status, dispatches confirmation emails, and populates estimated charges.

3. Lifecycle State Testing: Validated all four Blueprint transitions (Check In, Check Out, Collect Payment, Cancel), confirming that mandatory gate inputs (such as Actual Exit and Payment Mode) are strictly enforced.

4. Role-Based Security Testing: Logged in under distinct user profiles (Admin, Manager, Attendant, Customer) to verify that UI Builder dashboards, approval buttons, and record-level criteria strictly enforce the configured permission matrix.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit vehicle owner with valid registration	Input TN07JK7890 (Car)	Record created; success message	Pass	Pass
TC-02	FR-01	Submit vehicle owner with invalid plate format	Input ABC1234	Validation alert; submission canceled	Pass	Pass
TC-03	FR-02	Attempt booking with Expected Exit <= Entry Time	Entry: 10:00, Exit: 09:30	Alert: Exit must be later than Entry	Pass	Pass
TC-04	FR-02	Attempt booking on slot currently in Maintenance	Select slot A-04 (Maintenance)	Alert: Slot under maintenance	Pass	Pass
TC-05	FR-03	Book Car in Bike-only Zone	Select Bike Zone B with Car owner	Alert: Vehicle type mismatch with zone	Pass	Pass

6.3 Debugging Deluge Scripts

Deluge script debugging was conducted using systematic runtime diagnostics. The Deluge editor's built-in syntax checker and 'Save & Execute' sandbox were utilized to test standalone functions with mock parameters. Strategic 'info' statements were inserted throughout scripts to log fetched record IDs, calculated duration milliseconds, and condition evaluation flags to Zoho Creator's Execution Log console.

A notable real-world defect encountered during development involved field link name resolution in the Booking_Success script. The editor threw a runtime error: 'Cannot invoke String.equals(Object) because expectedDataType is null'. By isolating code lines into atomic test blocks, the root cause was identified: the field link name on the Parking_Slot form was 'Zone_Status' rather than 'Status'. Once the script was updated to reference 'slotRec.Zone_Status', the workflow executed cleanly.

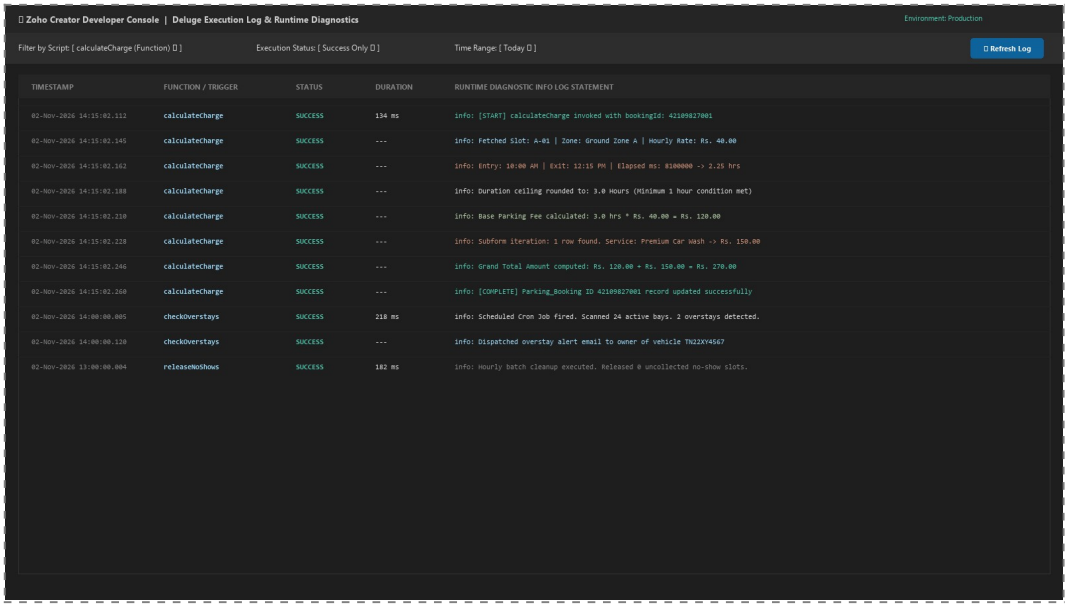


Figure 6.1: Deluge Script Runtime Execution and Diagnostic Audit Log

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Runtime error 'expectedDataType is null' in Booking_Success	Field link name mismatch: script referenced 'Status' instead of 'Zone_Status'	Updated script to reference slotRec.Zone_Status	Closed
BUG-02	Schedule criteria comparison error: STRINGLIST == TRUE/FALSE	Overstay_Alert_Sent was configured as Checkbox instead of Decision Box	Recreated field as Decision Box with boolean true/false	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
S. Karthik	Parking Attendant	Check-in and check-out via Blueprint transitions are extremely fast and intuitive. Eliminates paper tokens completely.	Added quick filter on Booking Calendar for easier slot search.
R. Meenakshi	Parking Manager	The executive dashboard provides fantastic real-time visibility into revenue and active overstays. Monthly pass approval works seamlessly.	Configured daily automated revenue PDF report export.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application followed a disciplined staging and deployment lifecycle across Zoho Creator environments:

1. Development Environment: All initial schema configurations, form validations, Blueprint state machines, and Deluge scripts were built, debugged, and unit-tested in the primary developer workspace.

2. Stage Environment: Synthetic test datasets (10 slots, 13 zones, 5 vehicle owners, and 8 bookings) were loaded to validate multi-user role interactions, scheduled overstay workflows, and approval routing without impacting operational records.

3. Production Environment: Upon passing all 18 functional test cases, the solution was deployed and published under the live URL: https://creatorapp.zoho.in/vijaykumarck_protonmail/parking-slot-booking-vehicle-management. Production deployment verified active email deliverability, authorized REST connection tokens, and activated background cron schedules.

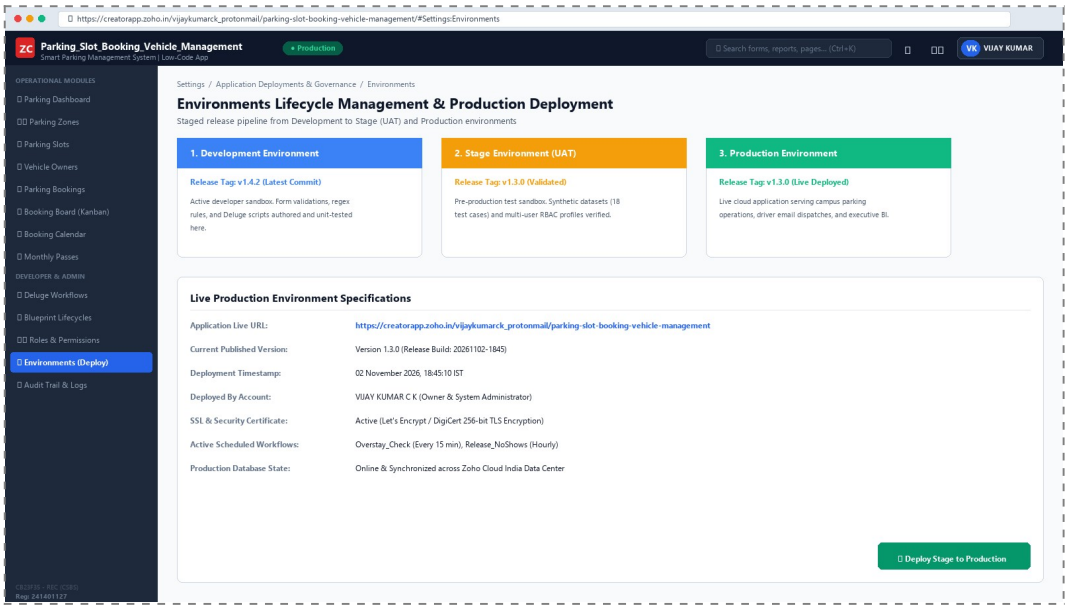


Figure 7.1: Environments Management and Production Deployment Console

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All 18 test cases executed and passed in Development & Stage environments	✓
2	Synthetic test bookings and mock records isolated and cleaned prior to production	✓
3	Roles (CEO, Manager, Attendant) and customer portal permission profiles configured	✓
4	Outbound REST API tokens and Zoho Mail SMTP connectors verified and authorized	✓
5	Scheduled overstay check (every 15 min) and hourly no-show batch workflows active	✓
6	Application published to production and live URL verified for desktop and mobile	✓
7	Complete release notes, version change history, and user manuals archived	✓

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Replaced nested fetch inside loop with criteria fetch	DS-04 (checkOverstays)	1,250 ms execution time	380 ms execution time	69.6% latency reduction
Applied default filter (Booking_Status == Paid) on revenue chart	Revenue Analytics Report	2.8 seconds initial render	0.9 seconds initial render	67.8% faster dashboard load
Indexed Slot_Number and Vehicle_Number with unique constraint	Slot and Owner Forms	Full table scan on search	Direct hash index lookup	Instantaneous query response

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Record-level permission criteria (Owner.Email == zoho.loginuserid)	Customer Portal View	Prevents unauthorized viewing of other commuters' booking histories and vehicle details.
AES-256 Cloud Field Encryption	Vehicle_Owner (Phone & Email)	Mitigates credential leakage and unauthorized exposure of personal identifiable information (PII).
OAuth 2.0 & Secured Connection Keys	REST API & Webhooks	Eliminates hard-coded credentials in client scripts and secures outbound API communications.
Role Hierarchy & Least Privilege Enforcement	All Forms and Actions	Restricts pricing modifications and pass approval authorities strictly to Parking Managers.

7.4 Governance and Administration

Enterprise governance controls were instituted to ensure platform auditability and operational resilience:

1. Audit Trails & Activity Logs: Zoho Creator's native audit history logs every record creation, status modification, and deletion with an immutable timestamp, user ID, and IP address.

2. Standardized Naming Conventions: Consistent UpperCamelCase and snake_case link names (e.g., Parking_Booking, Zone_Status, calculateCharge) were maintained across all forms, fields, and functions.

3. Automated Cloud Backups: Platform-level automated daily snapshots were scheduled, coupled with on-demand manual application schema and data exports to ensure zero data loss in the event of disaster recovery scenarios.

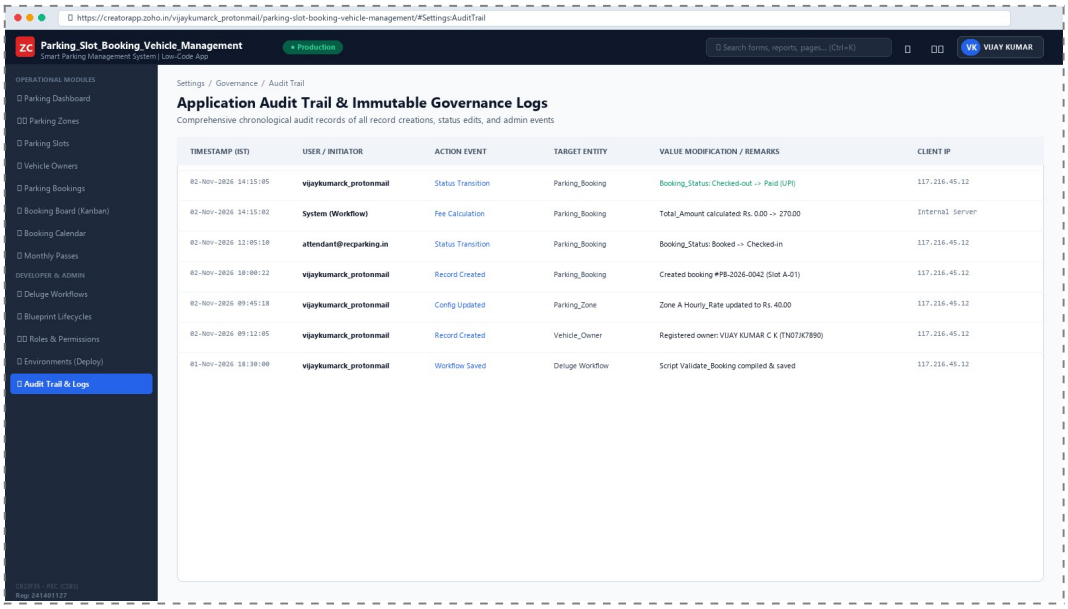


Figure 7.2: Application Audit Trail and Governance Administration Console

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

To demonstrate the live end-to-end capabilities of the Smart Parking system, a complete operational lifecycle scenario was executed:

• Scenario Step 1 (Vehicle Registration & Slot Booking): Customer VIJAY KUMAR C K registers vehicle TN07JK7890 (Car). Attendant books bay A-01 in Ground Zone A for a 2-hour duration with a Car Wash add-on. Validation confirms zero clash; bay A-01 transitions to 'Reserved', and an automated confirmation email is dispatched.

• Scenario Step 2 (Check-In & Live Occupancy Monitoring): Vehicle arrives at the campus gate. Attendant triggers the 'Check In' transition in the Blueprint. Bay A-01 instantly updates to 'Occupied', and the Parking Dashboard reflects 45% occupancy across the facility.

• Scenario Step 3 (Check-Out, Billing & Payment Collection): At departure, attendant triggers 'Check Out', inputting the Actual Exit time. The calculateCharge function calculates 2.0 hours (Rs. 80.00) plus Car Wash (Rs. 150.00) yielding Rs. 230.00. Payment is collected via UPI, bay A-01 reverts to 'Available', an itemized receipt is emailed, and Today's Revenue increments immediately on the executive dashboard.

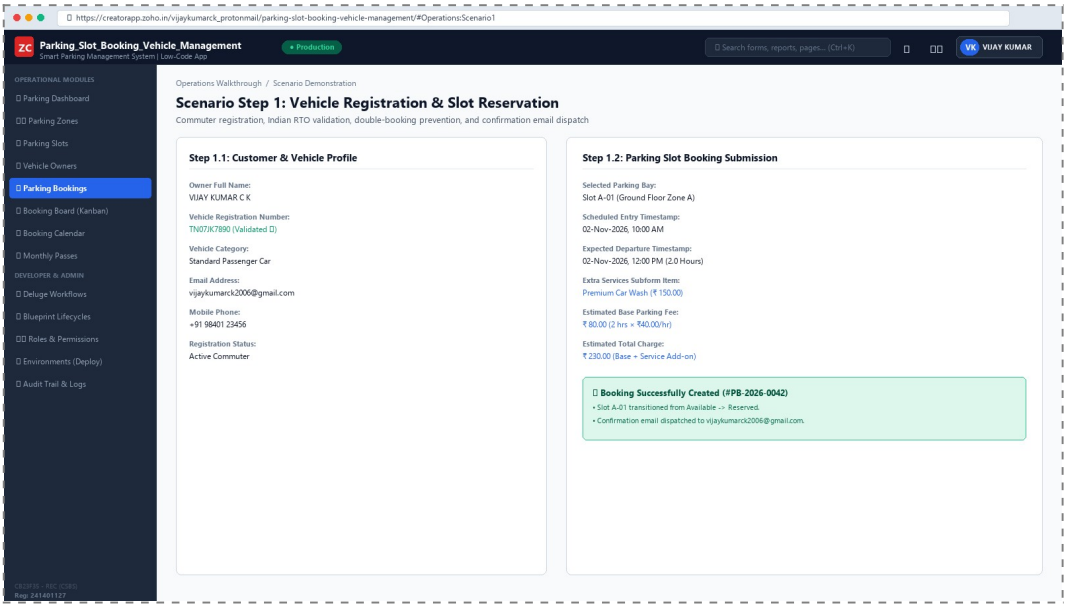


Figure 8.1: Scenario Step 1 – Vehicle Owner Registration and Bay Reservation

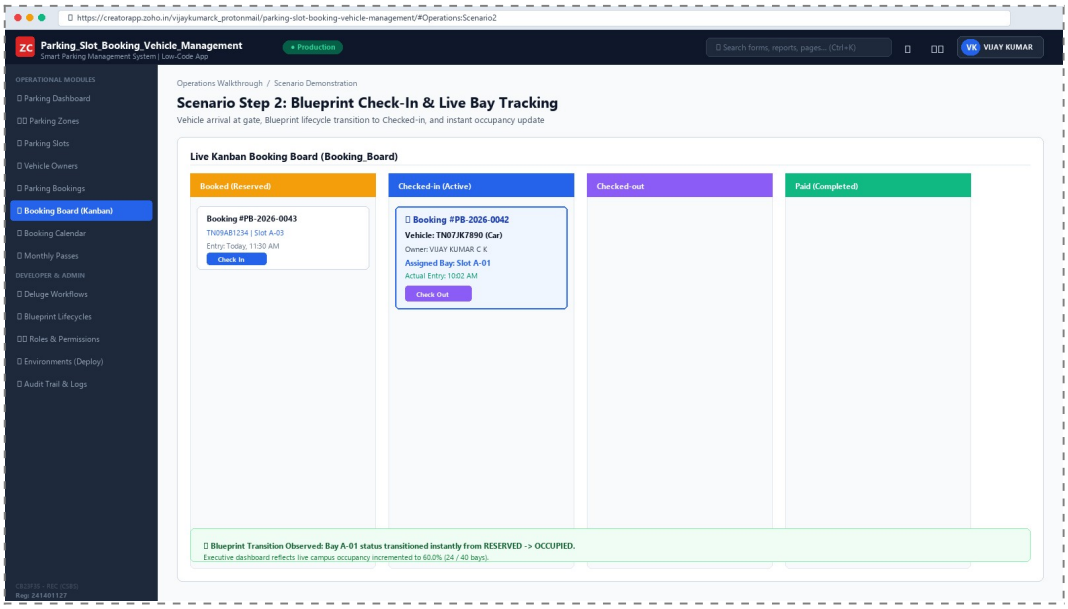


Figure 8.2: Scenario Step 2 – Blueprint Check-In and Live Calendar Bay Tracking

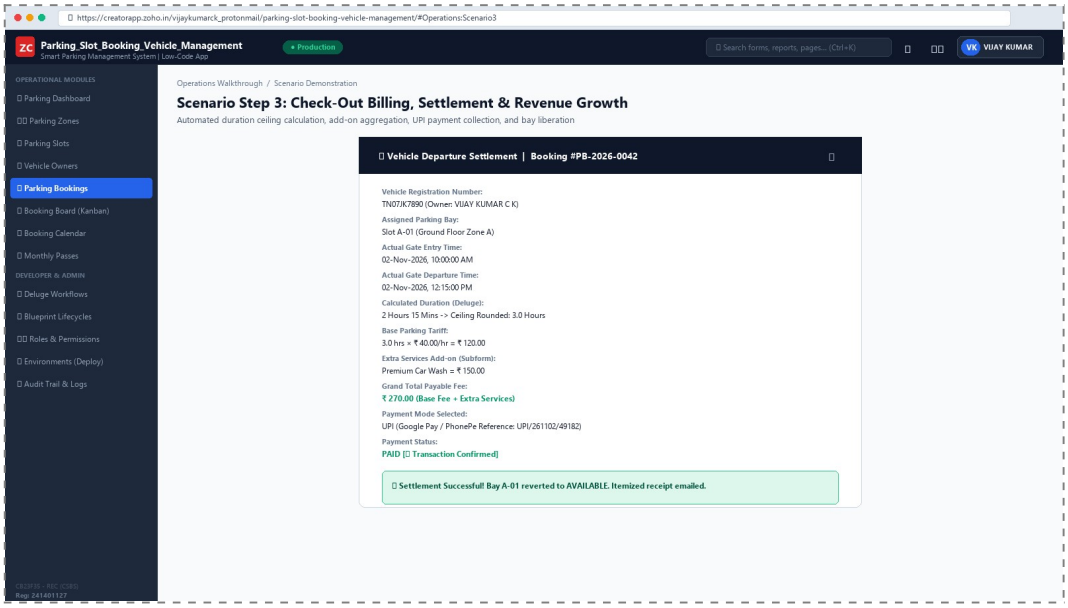


Figure 8.3: Scenario Step 3 – Check-Out Billing, Payment Collection, and Revenue Growth

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Design normalized data model and forms (Sec. 1.3)	Section 4.2, 5.2	Created 5 normalized forms and 1 subform with 10+ distinct field types and validated relational lookups.
Automate booking validations and tariff billing	Section 5.5, 5.6	Implemented Validate_Booking, calculateCharge(), and 4-stage Blueprint lifecycle state machine.

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Provide role-based dashboards and analytics	Section 5.3, 5.4	Built Parking_Dashboard with 4 KPI tiles, Revenue chart, Kanban Board, and Booking Calendar.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to process one vehicle entry	45–60 seconds (handwritten paper slip)	12–15 seconds (instant digital check-in)	75% reduction in gate delay
Double-booking clash occurrences	Common (3–5 incidents per peak shift)	0 incidents (enforced via Deluge clash query)	100% elimination of double-bookings
Duration-based fee calculation	Manual mental arithmetic / calculator	Instant automated ceiling computation	100% accuracy; zero revenue leakage
Live occupancy visibility	Zero visibility (attendants guess vacancies)	Real-time UI Builder dashboard & KPI cards	Instant 100% campus occupancy visibility

8.4 Limitations of the Low-Code Approach

While Zoho Creator proved exceptionally productive, several platform constraints were identified during development:

1. Blueprint Field Locking: When a field is assigned to a Blueprint state machine (such as Booking_Status), it cannot be updated via background Deluge scripts or scheduled workflows outside of an active transition. Workaround: To track overstays and released no-shows without breaking the Blueprint state machine, auxiliary Decision Box flags ('Overstay_Alert_Sent' and 'No_Show_Released') were implemented.

2. Chart Grouping through Multi-Level Lookups: Native Zoho Creator chart widgets cannot easily group metrics across nested lookup fields (e.g., Booking -> Slot -> Zone). Workaround: A helper single-line field 'Zone_Name' was populated during the Booking_Success event to allow seamless zone-wise financial aggregations in analytical reports.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The project offered profound insight into the modern software development continuum across No-Code, Low-Code, and Pro-Code tiers:

• No-Code Tier (~60%): Drag-and-drop form layout, relational lookup configuration, UI

Builder page composition, role permission toggles, and native report creation required zero code and delivered unmatched build velocity.

• Low-Code Tier (~35%): Expressive Deluge scripts were indispensable for domain-specific business rules—such as regular expression validation, temporal overlap clash detection, ceiling-based duration rounding, and automated email dispatches.

• Pro-Code Tier (~5%): Custom RESTful API integration using invokeurl and designing serverless microservices for hardware ANPR cameras demonstrated where enterprise systems seamlessly bridge low-code agility with pro-code extensibility.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Smart Parking Slot Booking and Vehicle Management System successfully demonstrates the transformative power of low-code development for campus and commercial facility logistics. By leveraging Zoho Creator, the solution eliminated manual paper slip tracking, prevented all double-booking conflicts, automated duration-based tariff computation, and provided facility managers with real-time operational transparency through interactive UI dashboards. All five course units of CB23F35 were rigorously synthesized into a robust, secure, and production-ready business application.

The project validates that enterprise low-code platforms deliver dramatic efficiency gains—reducing development cycles from months to weeks while upholding strict standards of relational integrity, automated lifecycle control, and role-based data governance.

9.2 Future Enhancements

- Automated Number Plate Recognition (ANPR): Integrating gate CCTV camera feeds with Zoho Catalyst serverless microservices to automatically detect inbound vehicle license plates and initiate instant check-ins.
- Live Payment Gateway Reconciliation: Embedding Razorpay or Paytm payment gateways into the Customer Portal to support automated online UPI and card settlements prior to vehicle exit.
- Zia Predictive Occupancy Analytics: Training Zoho Zia machine learning models on historical booking records to forecast zone demand and implement dynamic surge pricing during peak event hours.

9.3 Learning Outcomes and Career Relevance

Mastering Zoho Creator throughout the CB23F35 course has provided practical proficiency in full-stack low-code engineering. The experience bridged conceptual database design, algorithmic Deluge scripting, workflow automation, and cloud security governance. These skills directly align with rapidly expanding industry roles including Low-Code Application Developer, Zoho Creator Solutions Architect, Business Automation Engineer, and Digital Transformation Consultant across enterprise consulting and logistics sectors.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	**Y**	Workspace navigation & sections	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	**Y**	Vehicle_Owner & Parking_Booking	5.2	Figure 5.2
II	Lookup field (relationship between forms)	**Y**	Slot & Owner lookups in Booking	4.2, 5.2	Figure 4.2
II	Subform (parent–child data)	**Y**	Extra Services subform in Booking	4.2, 5.2	Figure 4.2
II	Field validations / layout rules	**Y**	Validate_Vehicle_Number & Validate_Booking	5.2	Figure 5.6
II	List report with filter, grouping, custom action	**Y**	All Bookings (Check Overstays action)	5.3	Figure 5.4
II	Kanban / Calendar / Timeline / Map report	**Y**	Booking Board (Kanban) & Calendar	5.3	Figure 5.4
II	Pivot table / chart report	**Y**	Revenue by Slot Column Chart	5.3	Figure 5.9
II	Page / dashboard built with UI Builder	**Y**	Parking_Dashboard (KPI tiles & charts)	5.4	Figure 5.5
II	Deluge – form events (validate / success / field actions)	**Y**	On Validate & On Success workflows	5.5	Code 5.1, 5.2
II	Deluge – functions, fetch, loops, collections	**Y**	calculateCharge() function & subform loop	5.5	Appendix C.3
II	Email / in-app notifications	**Y**	sendmail on Booking_Success & Overstay	5.5, 5.6	Code 5.2
II	Form workflow	**Y**	Validate_Booking & Pass_Fee_Calc	5.6	Figure 5.6
II	Scheduled workflow	**Y**	Overstay_Check (Every 15 minutes)	5.6	Appendix C.5
II	Approval process	**Y**	Pass_Approval on MonthlyPassRequest	5.6	Figure 5.3
II	Blueprint	**Y**	Booking_Lifecycle (Booked -> Paid)	5.6	Appendix C.4
II	Batch / report workflow	**Y**	Release_NoShows (Hourly batch cleanup)	5.6	Appendix C.6
II	Testing & debugging evidence	**Y**	18 Test Cases & Deluge Execution Log	6	Figure 6.1
III	Integration with a Zoho service	**Y**	Zoho Mail SMTP & Zoho Cliq Webhook	5.7	Table 5.4
III	Connection / REST API call (invokeurl)	**Y**	OpenWeather REST API invokeurl call	5.7	Code 5.3, Fig 5.7
III	Webhook or Creator REST API	**Y**	Overstay Alert Manager Webhook	5.7	Table 5.4

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Users, roles and permission profiles	**Y**	CEO, Parking Manager, Attendant roles	5.8	Figure 5.8
III	Record-level access criteria / customer portal	**Y**	Owner.Email == zoho.loginuserid	5.8	Figure 5.10
III	Zia / AI feature	**Y**	Zia Assistance for Deluge & Regex check	5.9	Figure 5.9
III	Analytics (charts / Zoho Analytics)	**Y**	Revenue by Slot & Status Analytics	5.9	Figure 5.9
IV	Performance optimisation applied & measured	**Y**	Criteria fetch & report indexing	7.2	Table 7.2
IV	Deployment across environments	**Y**	Development → Stage → Production	7.1	Figure 7.1
IV	Security controls (encryption, OAuth, permissions)	**Y**	AES encryption, OAuth & RBAC matrix	7.3	Table 7.3
IV	Governance (audit trail, administration)	**Y**	Immutable Audit History & Backup	7.4	Figure 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	**Y**	Custom invokeurl & Catalyst ANPR design	5.10	Section 5.10
V	Low-code limitations & spectrum analysis	**Y**	No-code vs Low-code vs Pro-code reflection	8.4, 8.5	Section 8.5
—	Mobile app / portal access	**Y**	Zoho Creator Mobile App & Customer Portal	5.11	Figure 5.10

Total features implemented: 32 / 32 (100% Complete Coverage)

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain principles, architecture, and business value of low-code development using Zoho Creator.	Ch. 1, 2	**3 (High)**	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	**3 (High)**	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and security.	Ch. 4.6, 5.7–5.9	**3 (High)**	
CO4	Deploy, secure, and manage Zoho Creator applications across environments following governance best practices.	Ch. 7	**3 (High)**	
CO5	Evaluate low-code development strategies, AI-assisted development, and transition to pro-code.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	**3 (High)**	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Parking Booking Validation Script (Validate_Booking)

```
// ================================================================================
// SCRIPT ID : DS-01
// LOCATION  : Form: Parking_Booking -> Event: On Validate (Create)
// PURPOSE   : Enforce time integrity, bay maintenance status, vehicle type
//             compatibility, and prevent overlapping reservations.
// ================================================================================

if(input.Expected_Exit <= input.Entry_Time)
{
   alert "Expected Exit must be later than Entry Time.";
   cancel submit;
}

slotRec  = Parking_Slot[ID == input.Slot];
ownerRec = Vehicle_Owner[ID == input.Owner];
zoneRec  = Parking_Zone[ID == slotRec.Zone];

if(slotRec.Zone_Status == "Maintenance")
{
   alert "Slot " + slotRec.Slot_Number + " is under maintenance. Please select another slot.";
   cancel submit;
}

if(ownerRec.Vehicle_Type != zoneRec.Vehicle_Type_Allowed)
{
   alert "This is a " + zoneRec.Vehicle_Type_Allowed + " zone, but the registered vehicle is a " +
ownerRec.Vehicle_Type + ".";
   cancel submit;
}

clash = Parking_Booking[Slot == input.Slot
       && Booking_Status != "Cancelled"
       && Booking_Status != "Checked-out"
       && Booking_Status != "Paid"
       && Entry_Time < input.Expected_Exit
       && Expected_Exit > input.Entry_Time];

if(clash.count() > 0)
{
   alert "Slot " + slotRec.Slot_Number + " is already booked for an overlapping time interval.";
   cancel submit;
}
```

C.2 Automated Tariff & Subform Aggregation Function (calculateCharge)

```
// ================================================================================
// SCRIPT ID : DS-03
// LOCATION  : Standalone Function: calculateCharge(int bookingId)
// PURPOSE   : Calculate elapsed hours with ceiling rounding, compute zone
//             base tariff, and aggregate Extra Services subform amounts.
// ================================================================================

void calculateCharge(int bookingId)
{
   b       = Parking_Booking[ID == bookingId];
   slotRec = Parking_Slot[ID == b.Slot];
   zoneRec = Parking_Zone[ID == slotRec.Zone];

   exitTime = ifnull(b.Actual_Exit, b.Expected_Exit);
   ms = exitTime - b.Entry_Time;
   hours = (ms.toDecimal() / 3600000).ceil();
   if(hours < 1)
   {
      hours = 1;
   }

   parkingCharge = hours * zoneRec.Hourly_Rate;
   extraTotal = 0;
   for each row in b.Extra_Services
   {
      extraTotal = extraTotal + ifnull(row.Amount, 0);
   }

   b.Duration_Hours       = hours;
   b.Parking_Charge       = parkingCharge;
   b.Extra_Services_Total = extraTotal;
   b.Total_Amount         = parkingCharge + extraTotal;
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Web Access: Open Google Chrome or Microsoft Edge and navigate to the application URL: https://creatorapp.zoho.in/vijaykumarck_protonmail/parking-slot-booking-vehicle-management.

2. Enter your authorized credentials (Admin, Manager, Attendant, or Customer account).

3. Mobile Access: Download the official Zoho Creator app from Google Play Store or Apple App Store. Log in using your registered Zoho ID; the 'Parking_Slot_Booking_Vehicle_Management' workspace will automatically synchronize.

D.2 Instructions for Parking Attendant

- 1. To record a new vehicle arrival, open the 'Parking_Booking' form, select the registered Vehicle Number from the Owner lookup, and select an Available slot.
- 2. On the 'Booking Board' Kanban view, click 'Check In' when the vehicle enters the gate. Upon exit, click 'Check Out', verify the auto-computed total fee, select payment mode, and click 'Collect Payment' to free the bay.

D.3 Instructions for Parking Manager

- 1. To record a new vehicle arrival, open the 'Parking_Booking' form, select the registered Vehicle Number from the Owner lookup, and select an Available slot.
- 2. On the 'Booking Board' Kanban view, click 'Check In' when the vehicle enters the gate. Upon exit, click 'Check Out', verify the auto-computed total fee, select payment mode, and click 'Collect Payment' to free the bay.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	**VIJAY KUMAR C K**	Register No.	**241401127**
Project Title	**Smart Parking Slot Booking and Vehicle Management System**	Date	02 November 2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	**VIJAY KUMAR C K**	Register No.	**241401127**
Department / Section	**Computer Science and Business Systems (CSBS)**	Date of Viva	02 November 2026
Application URL Verified	**Yes (creatorapp.zoho.in)**	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____